

Richie Nguyen

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Education

California State University, Fullerton

Fullerton, CA

Bachelor of Science in Computer Science, GPA: 3.7

May 2027

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Cybersecurity Principles, Applied AI, Operating System Concepts, Network Security, File Structure and Database

Experience

Dagster–Pandera — CTI CodeDay Labs

Open-Source Software Engineering Mentorship

Jun 2025–Aug 2025

- Diagnosed and fixed a schema propagation defect in Pandera affecting 3 validation code paths; authored PR #2111 (~250 LOC across 6 files), merged by maintainers into a library with 3.9k+ GitHub stars and 2M+ monthly downloads.
- Wrote 20+ pytest unit and integration tests in Python, Pandas, and Pydantic covering nested DataFrame schema edge cases, raising module-level coverage from 71% to 94% and eliminating a class of silent validation failures.

Eclipse — CodeDay Labs

Software Engineering Mentorship

Oct 2024–Nov 2024

- Partnered with mentor Varun Chawla (VISA) on enhancing search functionality in the Eclipse Platform, a Java codebase exceeding 2M lines across 1,000+ plug-in projects.
- Implemented a Java-based path filter allowing developers to exclude directories by glob pattern, cutting irrelevant search hits by ~60% on workspaces containing 10k+ build and dependency files.
- Shipped the feature with 12 JUnit test cases and incorporated feedback from 3 rounds of maintainer review before submission upstream.

Projects

DermCat – Facial Blemish Tracker / React, Three.js, MediaPipe, FastAPI, PostgreSQL, Docker

- Built an app that tracks blemishes across scans by pinning each spot to MediaPipe's 898-triangle face mesh, so positions stay accurate as the head turns.
- Kept all photos on the user's device by running face detection, skin masking, and redness detection in the browser, with each upload under 64 KB.
- Stored scan history in PostgreSQL under anonymous hashed IDs and displayed it as a 3D face overlay in Three.js.
- Set up GitHub Actions CI that tests database migrations both ways and runs the full Docker Compose stack.

Wiki for Neanderthals – One-Syllable Chatbot / React, FastAPI, PostgreSQL, Docker

- Designed a RAG pipeline that rewrites the question, finds matching articles with pgvector search, and filters the LLM's answer.
- Built a chatbot that answers questions from Simple English Wikipedia using only one-syllable words.
- Swapped long words for one-syllable synonyms using WordNet, with a live check that flags long words as the user types.

Frogodoro – Focus App / React, Tailwind CSS, Firebase Auth, Firestore, Vercel, Figma

- Designed wireframes in Figma and built a focus web app featuring customizable timers, lo-fi music, and a pixel-art frog.
- Added Firebase login with Firestore sync so settings and focus stats carry over across devices.
- Cut background load time by 91% (19.9s to 1.8s on 4G) and build size by 62% by compressing images, GIFs, and audio.

Skills & Certifications

Programming: Python, JavaScript, TypeScript, HTML, CSS, Swift, SQL

Frameworks: React.js, Next.js, FastAPI, Express.js, Tailwind CSS

Databases & Tools: PostgreSQL, SQLite, Firebase, Supabase, Git/GitHub, Docker, Vercel, CI/CD, Figma